

Chittagong University of Engineering & Technology
Department of Electronics & Telecommunication Engineering

ETE - 403

VLSI Technology

Comprehensive Previous Year Question Bank
All Unique Questions - Numerical Variants Included

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Program: B.Sc. Engineering -- Level 4, Term 1

Department: Electronics & Telecommunication Engineering

Institution: CUET, Chittagong, Bangladesh

Years: 2016 - 2017 - 2018 - 2019 - 2020 - 2022 - 2023 - 2024

Full Marks: 210 - Time: 3 Hours

Answer any THREE questions from each section. Use separate script for each section.

SECTION A -- MOS/CMOS Device Physics, Circuits & Design

* IC Fundamentals, Moore's Law & Technology Choice

Q1.	What is Moore's Law? Explain the integrated circuit trends over the years and how they correspond to Moore's Law.	[10]
Q2.	Explain the choice of technology used for VLSI implementation. Discuss about the choice of technology (BJT vs MOS) in VLSI.	[15]
Q3.	Even though CMOS logic requires more transistors than NMOS logic to implement a circuit, why is CMOS preferred?	[10]
Q4.	What is an integrated circuit? What benefits do they have over traditional circuits? What are the advantages of integrated circuit over conventional circuit?	[10]
Q5.	Briefly explain the top-down approach in VLSI design.	[15]
Q6.	Give a brief comparison between MOS and Bipolar technology. Which is suitable for VLSI design and why? Write down the comparison between BJT and CMOS technology.	[10]
Q7.	Compare between the various design styles used in VLSI. Give your opinion on which is suitable for which application.	[15]

* MOS Device Physics: Modes, Threshold Voltage & I-V Characteristics

Q8.	What is threshold voltage? Explain how threshold voltage is calculated for MOS and how it is affected by the 'body effect'. Write down the expression of the threshold voltage taking into account the body effect of MOS device.	[10]
Q9.	Explain Accumulation mode, Depletion mode, and Inversion mode of MOS device with neat sketches.	[15]
Q10.	Derive the equations for I_{ds} for NMOS, operating in resistive and saturation modes. Draw the final I-V curve. Also show that the bias condition at the transition between saturation and non-saturation state is $V_{GS} = V_{DS} + V_{tn}$.	[20]
Q11.	Why is the substrate in NMOS connected to ground? Explain with example. Why is a PMOS transistor pass 1's well but 0's poorly? Why does an NMOS transistor pass 0's well but 1's poorly?	[10]
Q12.	Explain different types of regions of NMOS transistor.	[10]
Q13.	Explain any three of the following short-channel and hot-carrier effects: (i) Punch through (ii) Hot carrier effect (iii) Body effect (iv) Short channel effect (v) Drain-Induced Barrier Lowering (DIBL)	[15]
Q14.	Explain different types of scaling models of MOS transistors. What types of scaling factors are needed to be considered to accommodate the scaling model?	[15]
Q15.	Show that, decreasing L is the best way to improve speed of a transistor.	[10]

* Numerical: Bias State, Drain Current & Operating Point

- Q16.** Determine the bias state for the three circuit conditions if $V_{tn} = 0.4$ V. The source voltage is always lower than the drain voltage in NMOS transistor. Circuit (a): $V_G = 1.9$ V, $V_D = 2.5$ V, $V_S = 0$ V. Circuit (b): $V_G = 2.2$ V, $V_D = 0.5$ V, $V_S = -2.3$ V. Circuit (c): $V_G = 0.9$ V, $V_D = 0.5$ V, $V_S = -2.5$ V. [10]

> Variant (changed values):

Determine the bias state for three NMOS circuit conditions if $V_{tn} = 0.4$ V. Circuit (i): $V_G = 3$ V, $V_{top} = 1.4$ V, $V_{bot} = 0$ V. Circuit (ii): $V_G = 4$ V, $V_{top} = 2.7$ V, $V_{bot} = 2$ V. Circuit (iii): $V_G = 0$ V (open), $V_{top} = 3$ V, $V_{bot} = 5$ V. [12]

- Q17.** Determine the bias state for the PMOS transistor where $V_{tp} = -0.4$ V. Circuit (a): $V_G = 0$ V, $V_D = 2.5$ V, $V_S = 0$ V. Circuit (b): $V_G = 1.3$ V, $V_D = -2.3$ V, $V_S = 1.5$ V. Circuit (c): $V_G = -2.5$ V, $V_D = 0$ V, $V_S = -1.1$ V. [12]

- Q18.** Find R_D for the given transistor in Fig. 1(c) just at non-saturated biased. Given: $K_n = 50$ μ A/V², $V_{tn} = 0.4$ V, $W/L = 10$. Circuit: $V_{DD} = 2.5$ V, $V_G = 1$ V. [10]

> Variant (changed values):

Find R_D just at non-saturated bias. Given: $K_n = 80$ μ A/V², $V_{tn} = 0.5$ V, $W/L = 8$. Circuit: $V_{DD} = 3.3$ V, $V_G = 1.5$ V. [10]

- Q19.** Find I_D and V_{DS} and verify the bias state consistency for $V_{tn} = 0.5$ V, $K_n = 25$ μ A/V², $W/L = 2$. Circuit: $V_{DD} = 2$ V, $R = 12$ K, $V_G = 1.5$ V. [15]

> Variant (changed values):

Find I_D and V_{DS} and verify the bias state for $V_{tn} = 0.5$ V, $K_n = 25$ μ A/V², $W/L = 2$. Circuit: $V_{DD} = 2$ V, $R = 12$ K, $V_G = 1.5$ V (same circuit -- verify saturation region). [10]

- Q20.** Calculate I_D and V_{DS} if $K_n = 100$ μ A/V², $V_{th} = 0.6$ V, and $W/L = 3$ for transistor M_1 . Circuit: $V_{DD} = 5$ V through $R = 15$ K, $V_G = 1.8$ V. [12]

> Variant (changed values):

Calculate I_D and V_{DS} if $K_n = 120$ μ A/V², $V_{th} = 0.5$ V, and $W/L = 4$. Circuit: $V_{DD} = 5$ V through $R = 10$ K, $V_G = 2.2$ V. [12]

- Q21.** Calculate I_D and V_{DS} for $V_{tp} = -1.0$ V, $K_p = 100$ μ A/V², and $W/L = 4$. Circuit: $V_{DD} = 5$ V, load = 200 ohm, $V_G = 1.5$ V. [10]

> Variant (changed values):

Calculate I_D and V_{DS} for $V_{tp} = -0.7$ V, $K_p = 80$ μ A/V², and $W/L = 3$. Circuit: $V_{DD} = 3.3$ V, load = 150 ohm, $V_G = 1.0$ V. [10]

- Q22.** What value of R_D will raise V_0 to half of the power supply voltage (i.e. $V_0 = 0.5 V_{DD}$)? Given: $V_{tp} = -0.7$ V, $K_p = 80$ μ A/V², $W/L = 5$. Circuit: $V_{DD} = 3.3$ V, $V_G = 0$ V. [10]

> Variant (changed values):

Find R_D to set $V_0 = 0.5 V_{DD}$. Given: $V_{tp} = -0.5$ V, $K_p = 60$ μ A/V², $W/L = 4$. $V_{DD} = 5$ V. [10]

- Q23.** Given: $K'_n = 140$ μ A/V², $V_{tn} = 0.7$ V, $V_{DD} = 3$ V, $K'_p = 60$ μ A/V², $V_{tp} = 0.7$ V. Find: (a) t_x size ratio so that $V_M = 1.5$ V. (b) V_M if t_x are the same size. [10]

> Variant (changed values):

Given: $K'_n = 100$ μ A/V², $V_{tn} = 0.6$ V, $V_{DD} = 5$ V, $K'_p = 40$ μ A/V², $V_{tp} = 0.8$ V. Find: (a) Size ratio so that $V_M = 2.5$ V. (b) V_M if transistors are the same size. [10]

- Q24.** A CMOS inverter has: $B_n = 0.5$ mA/V², $B_p = 0.3$ mA/V², $V_{DD} = 3$ V, $C_{out} = 25$ μ F. Find the rising time and falling time. [10]

> Variant (changed values):

A CMOS inverter has: $B_n = 0.8$ mA/V², $B_p = 0.4$ mA/V², $V_{DD} = 5$ V, $C_{out} = 50$ μ F. Find the rising time and falling time. [10]

- Q25.** A step waveform changing from V_{dd} to 0V is applied to a CMOS inverter with initial output 0V. PMOS and NMOS are treated as switches with ON resistances R_p and R_n respectively. (i) Express output voltage V_o and current i_s as a function of time t . (ii) Obtain the energy supplied from the power source after sufficient time. (iii) Obtain the energy consumed by MOSFETs. (iv) If 'supplied energy > consumed energy', where is the surplus energy stored? [16]

* NMOS/CMOS Inverters, Transfer Curves & Logic Circuits

- Q26.** Draw and explain the operating principle of an NMOS inverter with an NMOS depletion transistor load. Explore the edge time response of the gate and figure out the rise time and maximum operating frequency of the circuit. [25]
- Q27.** What do you understand by contention current? If there is no contention current in the CMOS process, why does the PMOS need to be double the width of an NMOS? In a CMOS process, the PMOS device has twice the width of an NMOS device. If this ratio is maintained, why is it called a ratioless design? [20]
- Q28.** For a CMOS inverter, draw and explain the I vs V_{out} curve. Find out the transistor characteristics and power curve for CMOS. Explain CMOS inverter DC transfer curve with different regions. [15]
- Q29.** Show that in static behavior of an inverter, the switching threshold voltage is: $V_M = (V_{DD} - |V_{tp}| + V_{tn} * \sqrt{Bn/Bp}) / (1 + \sqrt{Bn/Bp})$. Where the symbols have their usual meaning. Also explain the effect of transistor size on VTC. [15]
- Q30.** Derive the inversion voltage of a transistor. Define inversion voltage and calculate the inversion voltage of a transistor. [15]
- Q31.** Draw the cross-section of a CMOS inverter and explain its operation. Also draw and mention its important parts. Explain the operating principle of CMOS inverter and draw its cross section. [15]
- Q32.** What do you understand by pass transistor and a pass gate? Explain why pass gates are used instead of pass transistors. [10]
- Q33.** What is the special feature of BiCMOS inverter? Explain with neat sketch and its operation. Give the schematic diagram of a Bi-CMOS inverter and explain its operation and properties. [10]
- Q34.** What are the key characteristics of MOS dynamic circuits? What makes dynamic CMOS circuit faster than a static CMOS circuit? Explain the limitations of cascaded dynamic CMOS logic gates. What are the possible solutions? [18]
- Q35.** What is latch-up? Explain latch-up with cross-section of a CMOS inverter. How can you avoid latch-up? Describe the latch-up phenomenon in CMOS circuits. What are the preventive design strategies to avoid latch-up? [12]
- Q36.** Explain the following effects in CMOS/MOS devices: (i) Hot Carrier Effect, (ii) Short Channel Effect, (iii) Drain-Induced Barrier Lowering. [12]
- Q37.** How do you improve switching speed within a specific gate? Explain. [10]
- Q38.** Explain CMOS inverter with resistive load. [10]
- Q39.** Derive the expression of rise time and fall time of a CMOS inverter. [14]
- Q40.** What do you mean by ratioed and ratio-less design? [5]

* CMOS Logic Gates, Boolean Functions & Stick Diagrams

Q41. Draw the schematic diagram of an XOR gate using CMOS process.	[10]
Q42. Draw the CMOS logic function: $Y = A.(B+C)(D+E+F)$.	[13]
Q43. Draw a CMOS circuit diagram of the following Boolean functions: (i) $AB + (C+D)E$ (ii) $(A(B+C)D + E)'$ (iii) $F = A(B+C)(D+E)$ (iv) $Y = A + B(C+D)$	[15]
Q44. What is stick diagram? Why is stick diagram necessary? Draw the stick diagram of: (i) A two-input NAND gate, (ii) A NOR gate, (iii) The equation $Y = (A+B)+C$.	[12]
Q45. Draw the stick diagram of a 1-bit full adder.	[15]
Q46. Explain precharge. Evaluate dynamic logic with necessary diagram.	[10]
* Layout Design, Lambda Rules & VLSI Design Methodology	
Q47. What do you understand by layout design? With proper explanation, draw and explain the layout of a CMOS inverter.	[25]
Q48. What do you understand by Lambda-based design rules? Explain how this affects VLSI design.	[8]
Q49. What do you understand by the standard-cell approach? Explain its cell characteristics and draw a multi-row standard cell layout showing various wire routings. Explain why standard-cell approach is suitable for design automation.	[13]
Q50. Using a Y-chart, explain the top-down/bottom-up approach towards VLSI design and manufacturing in behavioral, structural, and physical domains. Explain the Y-diagram of VLSI design for top-down design and bottom-up fabrication approach.	[15]
Q51. What is the significance of simulation in HDL-based design? Explain simplistic view of Design Flow. Briefly explain the behavioral, physical, and structural domain of design representation.	[13]
Q52. Briefly describe the following: (i) Floor planning (ii) Partitioning (iii) Static Timing Analysis (iv) Clock and Power Routing	[12]
Q53. Explain the importance of physical verification. Also explain how it is achieved.	[10]
Q54. What are the various design approaches adapted for implementing digital design?	[15]

SECTION B -- IC Fabrication, Memory, HDL & Physical Design

* Crystal Growth, Wafer Preparation & Oxidation

Q55.	Explain the basic fabrication steps in IC manufacturing.	[10]
Q56.	Describe the Czochralski method to form a Si ingot that can be sliced into wafer. Describe the process of silicon ingot preparation from raw silicon.	[18]
Q57.	What do you mean by material preparation? Explain the procedure of producing an ingot.	[15]
Q58.	Why is wet oxidation rate higher than dry oxidation rate? Explain. What do you understand by oxidation and what affects growth rate? Compare between wet oxidation and dry oxidation.	[10]
Q59.	Why is oxidation very much essential in IC fabrication? Write down the factors affecting oxide growth rate. Write down the typical properties of oxide.	[12]
Q60.	Write down the several applications of SiO_2 . Explain masking properties of SiO_2 .	[10]
Q61.	Describe various types of oxide charge at Si- SiO_2 interface. Also explain how the charges can be reduced.	[10]
Q62.	Describe thermal oxidation technique.	[12]
Q63.	What is diffusion and ion implantation? Write down several applications of SiO_2 . Why is ion implantation preferred over diffusion for impurity doping? Briefly explain the ion implantation technique.	[15]
Q64.	Which would you choose between diffusion and ion implantation for: (i) creating a deep, heavily doped region, (ii) creating a very shallow, precisely controlled region, and why?	[10]
Q65.	Write short notes on ion implantation and diffusion.	[8]

* Epitaxial Growth: CVD, LPE, MBE, Homo & Heteroepitaxy

Q66.	What do you understand by epitaxial growth? Explain why it is used and its advantages. What is the basic difference between Homoepitaxy and Heteroepitaxy?	[16]
Q67.	Explain CVD, LPE, and MBE methods for performing epitaxy. Explain their benefits and drawbacks. Briefly describe different types of epitaxial growth techniques.	[15]
Q68.	Explain Vapour phase epitaxy and Molecular Beam Epitaxy.	[18]

* Photolithography, Etching & Wafer Cleaning

Q69.	Explain the typical photolithography process in detail. Draw the different steps involved in the photolithography process. With neat sketches, show the step-by-step lithography process for manufacturing an NMOS.	[13]
Q70.	Write down the difference between positive photoresist and negative photoresist. Define photo resist and compare positive and negative photoresist.	[7]
Q71.	Explain the concept of mask alignment and its importance in multi-layer fabrication.	[10]
Q72.	What is etching? Explain different types of etching techniques. Differentiate between wet and dry etching.	[12]

Q73.	What is plasma? How is it produced? Explain about induction plasma. Describe the plasma chemistry in etching and the effects of O ₂ and H ₂ addition on etching rate.	[13]
Q74.	Define hydrophilic surface and hydrophobic surface. Write down the basic difference between hydrophilic and hydrophobic surface type.	[7]
Q75.	Why is wafer cleaning necessary? Briefly describe the steps of IMEC clean. Explain RCA clean in wafer cleaning process. Explain IMEC and RCA cleaning processes.	[18]
Q76.	Write short notes on: (i) Metallization, (ii) Passivation.	[8]
Q77.	What are meant by the terms: (i) FEOL, (ii) Hydrophilic and hydrophobic surface, (iii) Hetero and homo epitaxy, (iv) Passivation, (v) Metallization.	[10]
* Memory Arrays: SRAM, DRAM & Memory Types		
Q78.	Explain the different types of memory arrays used in VLSI circuits.	[10]
Q79.	Why is SRAM preferred in cache memory, while DRAM is used in main memory?	[7]
Q80.	Explain the operating principle for a 6T SRAM cell along with read and write operations and cell stability conditions. With proper diagram, explain the entire operating principle of the 6T SRAM cell along with cell stability criterion.	[15]
Q81.	How does BiCMOS combine the benefits of BJT and CMOS technologies?	[5]
* HDL/VHDL: Coding, Simulation & Digital Design		
Q82.	What does VHDL stand for? Explain the reason for using VHDL. Why is VHDL important?	[10]
Q83.	Design a single bit half adder with carry and enable using VHDL language.	[12]
Q84.	Write a structural HDL code for a 2-bit full adder using two 1-bit adders.	[10]
Q85.	Write an HDL code for generating a 4-bit Ripple Carry Adder using 4, 1-bit adders.	[10]
Q86.	Write the DUT and Testbench HDL code for a 1-bit Full-Adder.	[10]
Q87.	Write down the complete VHDL code for a single bit half adder.	[7]
* PLA, FPGA, Placement, Routing & Physical Design		
Q88.	What do you understand by PLA and FPGA? Explain the building blocks of an FPGA. What is an FPGA? Briefly discuss its various parts.	[10]
Q89.	Explain the Y-diagram of VLSI design for top-down design and bottom-up fabrication approach. Using a Y-chart, explain the top-down/bottom-up approach towards VLSI design and manufacturing in behavioral, structural, and physical domains.	[15]
Q90.	Explain how the TimberWolf algorithm works. How do ripple move and chain move work in the case of placements?	[10]
Q91.	What do you understand by Global Routing? Explain the sequential vs hierarchical approach towards Global Routing.	[10]
Q92.	Explain Hadlock's algorithm and the concept of detour numbers.	[10]

- Q93.** What is grid routing? With proper diagram, explain how Lee's algorithm works along with its memory requirements, and how can it be further optimized for reducing running time. [10]
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- Q94.** What do you understand by DFT? Why is it important? [5]
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-- END OF QUESTION BANK --

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